

Semantic segmentation

Classification



CAT

Single Object

**Semantic
Segmentation**

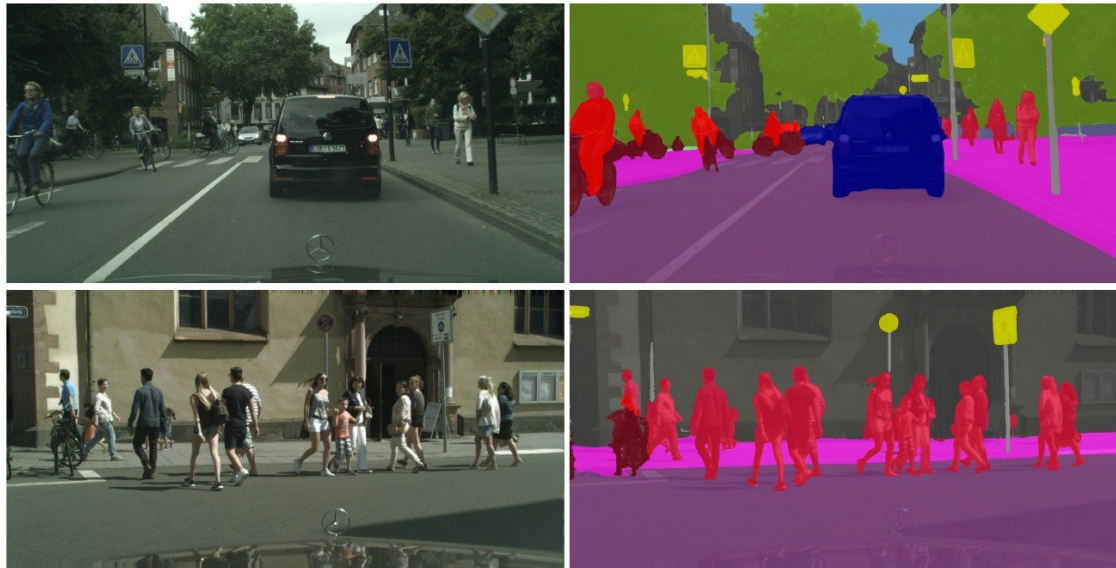


**GRASS, CAT,
TREE, SKY**

No objects, just pixels

Semantic Segmentation

- The goal is to predict the class for each pixel in the image.
- It is also called dense prediction.



- Applications:
 - Medical image diagnostics
 - Autonomous vehicles
 - Land cover classification

Segmentation Map

- The result of segmentation is the *segmentation map* where each pixel contains the corresponding class label.



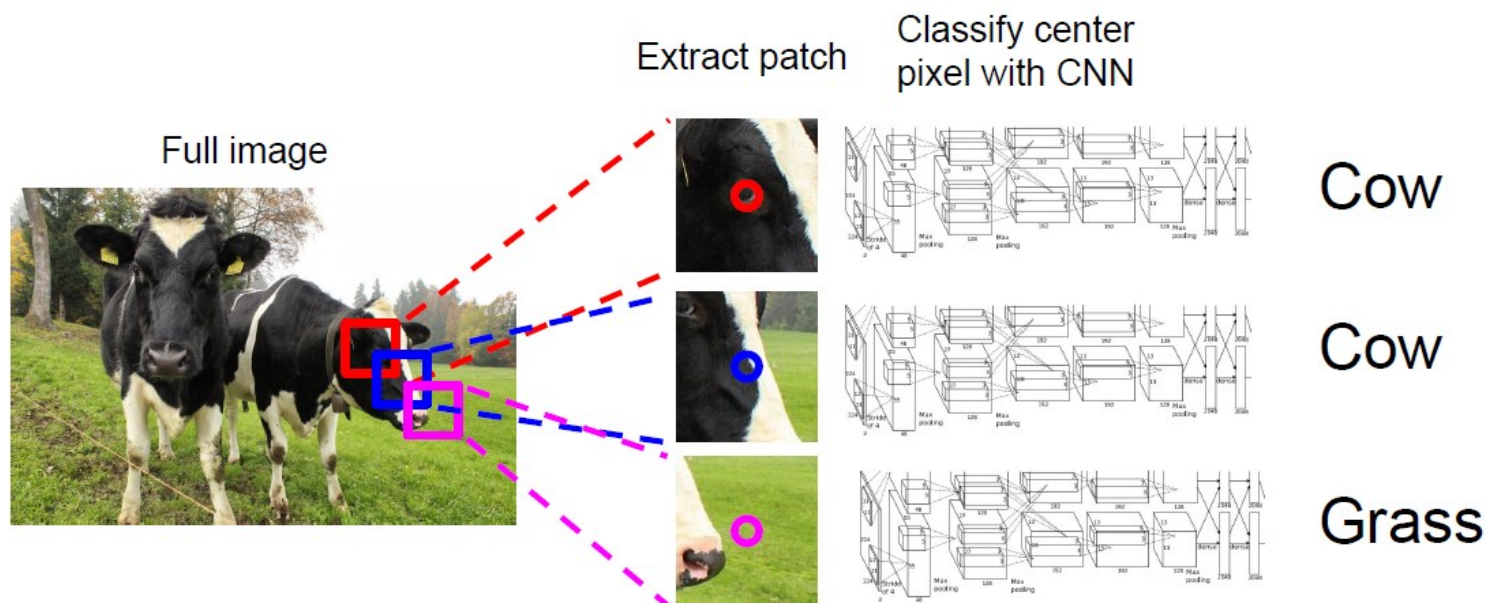
1: Person
2: Purse
3: Plants/Grass
4: Sidewalk
5: Building/Structures

3	3	3	3	3	3	3	3	3	3	3	3	3	5	5	5	5	5	5
3	3	3	3	3	3	3	3	3	3	3	3	3	5	5	5	5	5	5
3	3	3	3	3	3	1	1	3	3	3	3	3	5	5	5	5	5	5
3	3	3	3	3	1	1	1	1	3	3	3	3	5	5	5	5	5	5
3	3	3	3	3	3	1	1	3	3	3	3	5	5	5	5	5	5	5
5	5	3	3	3	3	1	1	3	3	5	5	5	5	5	5	5	5	5
4	4	3	4	1	1	1	1	1	1	4	4	4	5	5	5	5	5	5
4	4	3	4	1	1	1	1	1	1	4	4	4	4	4	5	5	5	5
4	4	4	1	1	1	1	1	1	1	1	4	4	4	4	4	4	4	4
3	3	3	1	1	1	1	1	1	1	1	4	4	4	4	4	4	4	4
3	3	3	1	2	2	1	1	1	1	1	4	4	4	4	4	4	4	4
3	3	3	1	2	2	1	1	1	1	1	4	4	4	4	4	4	4	4

- Loss function: Segmentation is equivalent to solving a classification problem for each pixel. We can use the sum of cross-entropy losses computed for each pixel.

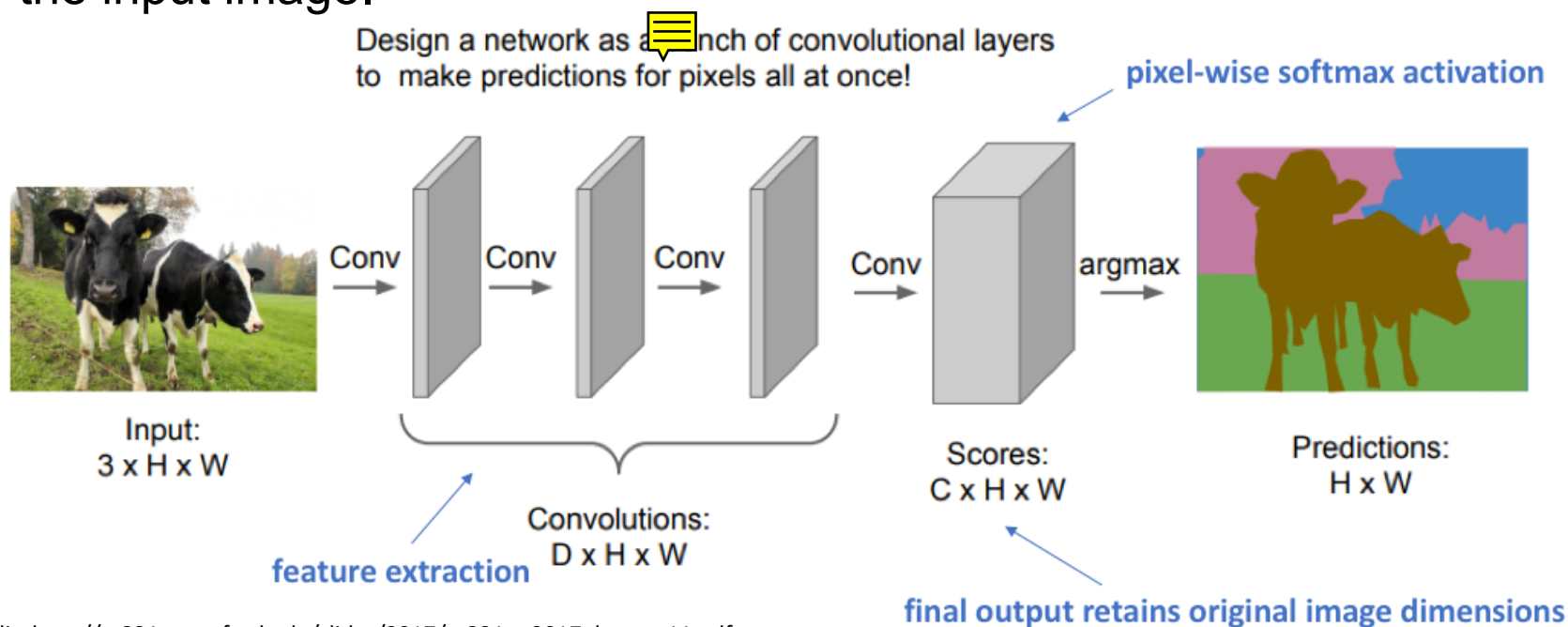
Semantic segmentation: sliding window

- It is highly inefficient: overlapping patches share the same features



Fully convolutional Network (FCN)

- The FCN does not include fully-connected layers but only 2d layers (e.g. convolutional layer, 2d pooling layer).
- It is an efficient alternative to applying the CNN in a sliding-window fashion.
- The output of the network should have the same spatial resolution as the input image.



Encoder/Decoder structure

- The architecture has two parts:

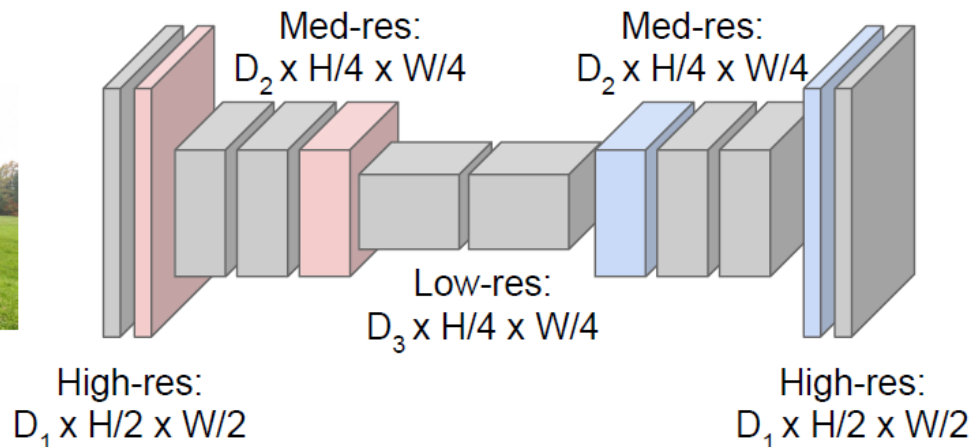


1. a Encoder: captures context and extracts high-level compact features. The spatial resolution is reduced and the number of feature maps is typically increased.
2. a Decoder: restores the output to the spatial resolution of the input image.

Design network as a bunch of convolutional layers, with **downsampling** and **upsampling** inside the network!



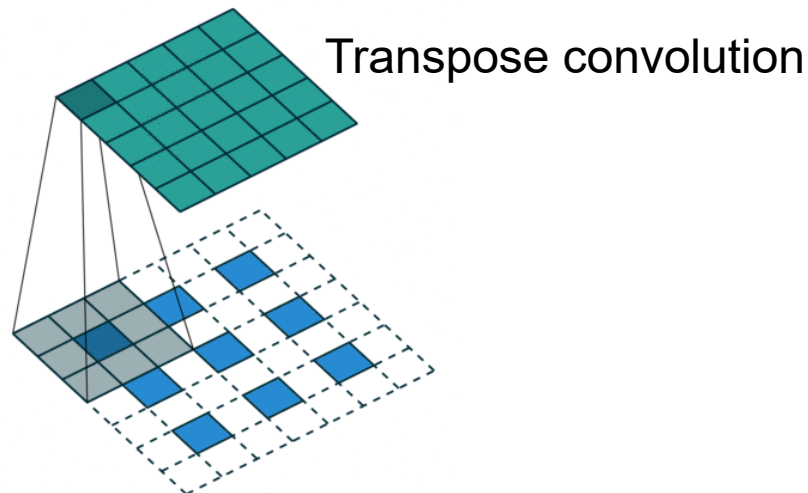
Input:
 $3 \times H \times W$



Predictions:
 $H \times W$

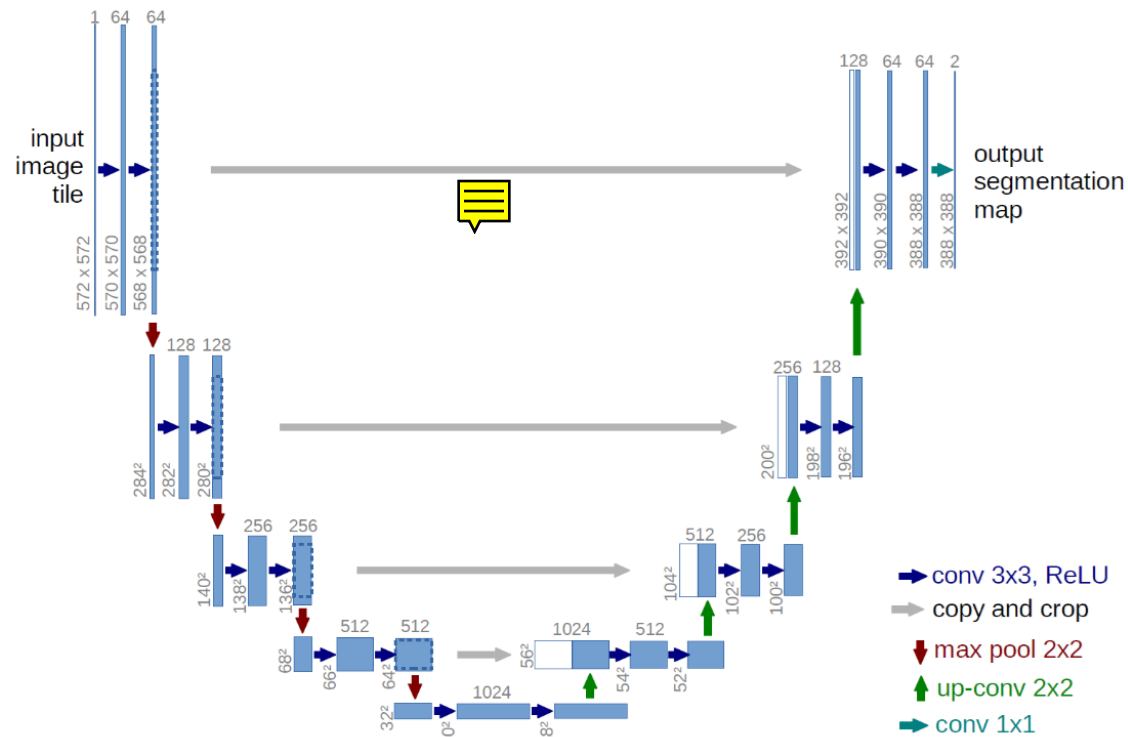
Methods for downsampling and upsampling

- The downsampling is always achieved through decimation in pooling or convolutional layers.
- The upsampling can be performed in different ways:
 - Transpose convolution (“fractionally strided convolution” or “deconvolution”).
 - Nearest Neighbor interpolation followed by a convolution
 - Unpooling



U-Net

- U-Net uses a symmetric architecture with skip connections between its encoder and decoder.
- Upsampling is performed using nearest-neighbor interpolation.



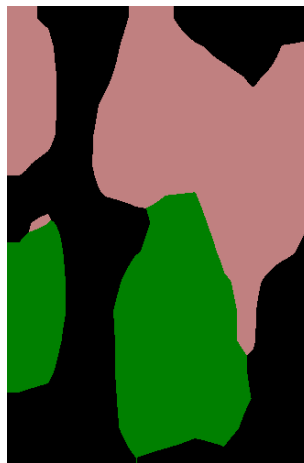
Skip connections

- Skip connections are used to recover finer-grained spatial detail.

Ground truth



Without skips



With skips



Segmentation metrics

Image



Desired map
(ground Truth)



Predicted map



Positive pixels (Positive): pixels belonging to the object we want to segment (car)

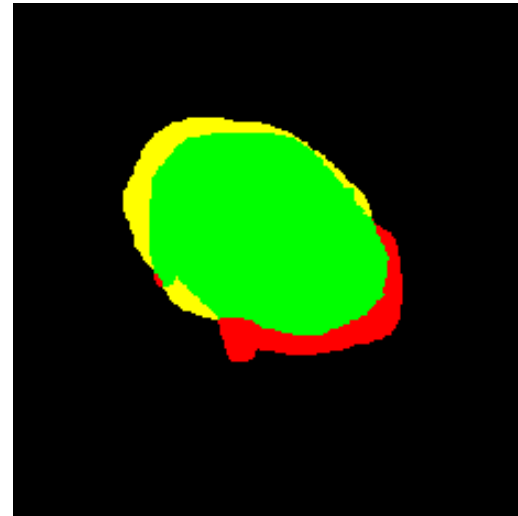
Negative pixels (Negative): pixels belonging to the background

Segmentation metrics

Image



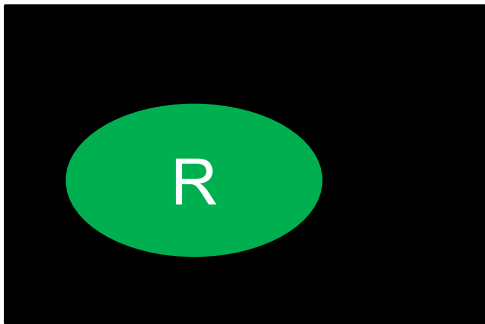
Desired map overlapping
the predicted map



-  Correctly identified Positive Pixels (True Positive)
-  Correctly identified Negative Pixels (True Negative)
-  Not Correctly identified Negative Pixels (False Positive)
-  Not Correctly identified Positive Pixels (False Negative)

Accuracy

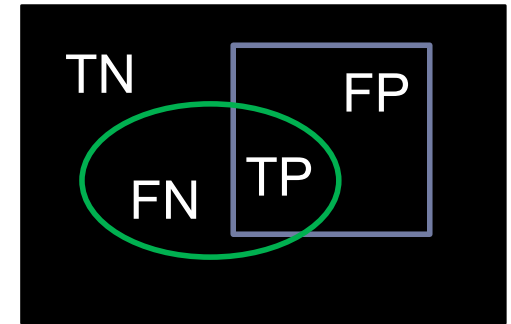
Desired Map



Predicted Map



Regions



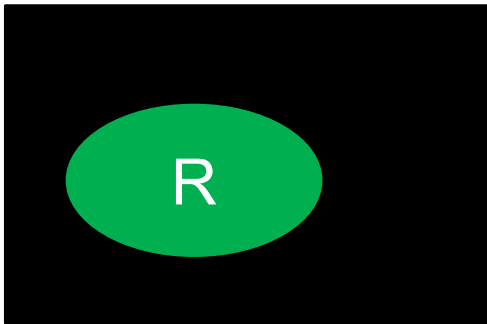
Accuracy

$$Acc = \frac{|TP| + |TN|}{|TP| + |TN| + |FP| + |FN|}$$

- The value is in the range $[0,1]$.
- It is equal to 1 if the maps are identical.
- It is strongly influenced by the size of the segmented region relative to the entire image.

Intersection over Union

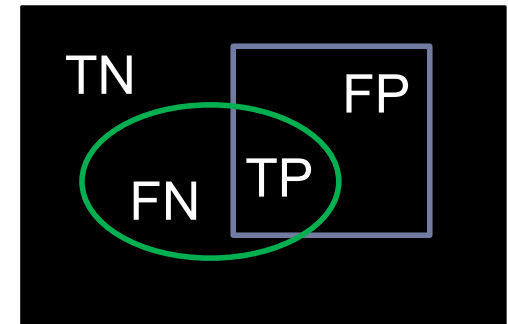
Desired Map



Predicted Map



Regions



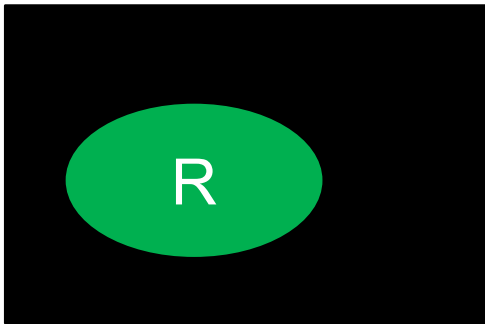
Intersection over Union
(Jaccard index)

$$IoU = \frac{|R \cap P|}{|R \cup P|} = \frac{|TP|}{|TP| + |FP| + |FN|}$$

- The value is in the range $[0,1]$. 
- It is equal to 1 if the maps are identical.
- It is not influenced by the size of the segmented region relative to the entire image.

F-measure

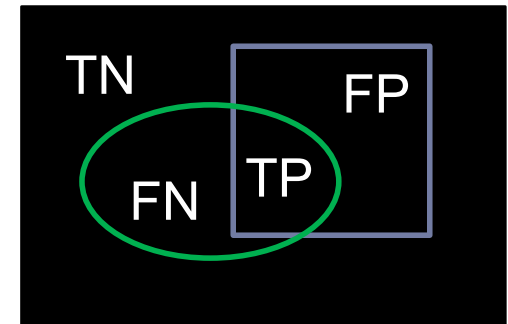
Desired Map



Predicted Map



Regions



F-measure
(Dice coefficient)

$$F1 = \frac{|R \cap P|}{(|R| + |P|)/2} = \frac{2 * |TP|}{2 * |TP| + |FP| + |FN|}$$

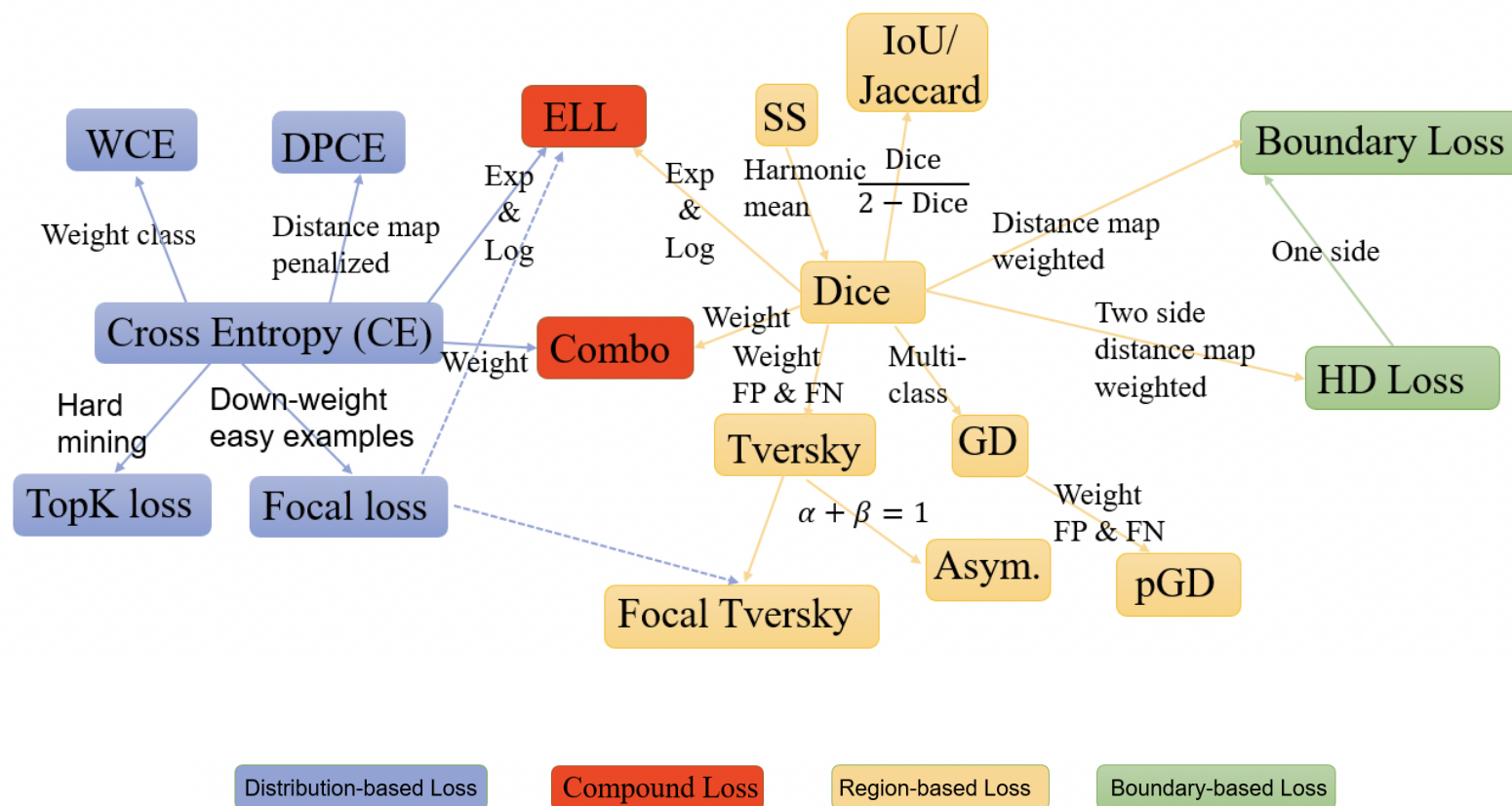
- The value is in the range [0,1].
- It is equal to 1 if the maps are identical.
- It is not influenced by the size of the segmented region relative to the entire image.

Loss functions for segmentation

- The pixel-wise cross-entropy loss is commonly used; it is equivalent to solving a classification problem for each pixel.
- However, when classes are unevenly represented in the image, training may be dominated by the most prevalent class.
- Countermeasures:
 - Pixel-wise losses: weight individual pixels in the loss function to counteract the imbalance (e.g., Weighted Cross-Entropy, Focal Loss).
 - Image-wise losses: compute an image-level overlap score between ground-truth and predicted regions (e.g., Jaccard Loss, Dice Loss).

Loss functions for segmentation

A survey by Jun Ma: <https://arxiv.org/pdf/2005.13449.pdf>



Focal Loss

- Focal Loss is a variant of Cross-Entropy loss
- It down-weights well-classified examples, focusing attention on harder ones.
- This is particularly helpful when training on images with many easy background pixels.

